

DATABASE PROJECT SUBMISSION WORKSHEET

Phase 2: Full Implementation & Documentation

Project Name	
Group Name	
Member 1	
Member 2	
Member 3	
Class / Batch	
Submission Date	
Instructor	Engr.Arifa

PROJECT SUBMISSION CHECKLIST

Tick each item once completed. Incomplete submissions will not be accepted.

#	Deliverable / Task	Marks / Status
☐	1. Cover Page with all member details filled	__ / 5
☐	2. Database Schema Diagram (ERD)	__ / 10
☐	3. Normalization — 1NF, 2NF, 3NF explained	__ / 15
☐	4. DDL Queries (CREATE, ALTER, DROP)	__ / 15
☐	5. DML Queries (INSERT, UPDATE, DELETE)	__ / 15
☐	6. DQL Queries — minimum 10 SELECT queries	__ / 20
☐	7. Database Backup (.sql dump file attached)	__ / 10
☐	8. Report Document (this worksheet completed)	__ / 10

PART 1 — DATABASE SCHEMA DIAGRAM

Draw or paste your Entity-Relationship Diagram (ERD) / Schema Diagram below. You may also attach a printed/digital copy of your diagram as an appendix.

Required elements in the diagram: All tables, Primary Keys (PK), Foreign Keys (FK), cardinality (1:1 / 1:N / M:N), and attribute data types.

[Attach / Draw ERD / Schema Diagram Here]

Table List — Write all table names in your database:

Brief description of each table and its purpose:

PART 2 — DATABASE NORMALIZATION

Demonstrate that your database satisfies all three normal forms. Answer each section for at least ONE table in your project (you may repeat for all tables).

2.1 — First Normal Form (1NF)

Rule: Each column must contain atomic (indivisible) values. No repeating groups.

State which table you are normalizing to 1NF:

Explain how your table satisfies 1NF (no multi-valued attributes, each cell has one value):

Example: The 'Orders' table has no repeating groups. Each product is stored in a separate row.

If modifications were required for 1NF, describe what changed:

2.2 — Second Normal Form (2NF)

Rule: Must be in 1NF AND every non-key attribute must be fully functionally dependent on the entire primary key (no partial dependency).

Identify the primary key for the table:

Explain why there is no partial dependency in your table:

Example: In 'OrderItems', every attribute depends on the composite key (order_id, product_id) — not just one part.

If partial dependencies existed, describe how you resolved them (e.g., split into new table):

2.3 — Third Normal Form (3NF)

Rule: Must be in 2NF AND no transitive dependencies (non-key attribute should not depend on another non-key attribute).

Check your table for transitive dependencies:

Explain how your table satisfies 3NF:

Example: 'city' does not depend on 'zip_code'; each attribute depends only on the primary key.

If transitive dependencies existed, describe the fix:

PART 3 — DATA DEFINITION LANGUAGE (DDL)

Write all DDL queries used to build your database structure. Include the actual SQL you executed.
 Required: *CREATE DATABASE*, *USE*, *CREATE TABLE (with constraints)*, *ALTER TABLE*, and *DROP TABLE*.

3.1 — Create Database & Use

Write your CREATE DATABASE and USE statement:

e.g. CREATE DATABASE school_db; USE school_db;

3.2 — Create Tables (with Constraints)

Write CREATE TABLE statements for ALL tables in your project. Include constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, DEFAULT, CHECK, AUTO_INCREMENT.

Table 1 — CREATE TABLE:

Table 2 — CREATE TABLE:

Table 3 — CREATE TABLE:

Table 4+ — CREATE TABLE (add more as needed):

3.3 — ALTER TABLE

Write at least 2 ALTER TABLE statements (ADD column, MODIFY column, ADD CONSTRAINT, DROP COLUMN, etc.)

ALTER TABLE statements:

e.g. ALTER TABLE users ADD COLUMN profile_pic VARCHAR(255);

3.4 — View Table Structure

Write DESCRIBE / SHOW COLUMNS statements for each table:

e.g. DESCRIBE users; SHOW COLUMNS FROM orders;

3.5 — DROP TABLE (Test Only)

Write a DROP TABLE statement (used only during testing — do NOT drop final tables):

e.g. DROP TABLE IF EXISTS temp_log;

PART 4 — DATA MANIPULATION LANGUAGE (DML)

Write all DML queries. You must INSERT at least 5 rows per table, perform at least 3 UPDATE operations, and at least 2 DELETE operations.

4.1 — INSERT Statements

Insert sample/test data into ALL tables. Use realistic values related to your project.

INSERT INTO — Table 1 (minimum 5 rows):

e.g. `INSERT INTO users (name, email, role) VALUES ('Ali', 'ali@email.com', 'admin');`

INSERT INTO — Table 2 (minimum 5 rows):

INSERT INTO — Table 3 (minimum 5 rows):

INSERT INTO — Remaining Tables:

4.2 — UPDATE Statements

Write at least 3 UPDATE queries. Use WHERE clause correctly to avoid updating all rows accidentally.

UPDATE statements (minimum 3):

e.g. UPDATE products SET price = 1500 WHERE product_id = 3;

4.3 — DELETE Statements

Write at least 2 DELETE queries. Always use WHERE clause.

DELETE statements (minimum 2):

e.g. DELETE FROM orders WHERE status = 'cancelled' AND order_date < '2023-01-01';

PART 5 — DATA QUERY LANGUAGE (DQL)

Write at least 10 SELECT queries that extract meaningful data from your database. Cover all the categories listed below.

5.1 — Basic SELECT Queries

Query 1: Select ALL records from any table:

e.g. `SELECT * FROM users;`

Query 2: Select SPECIFIC columns:

e.g. `SELECT name, email FROM users;`

Query 3: SELECT with WHERE condition:

e.g. `SELECT * FROM products WHERE category = 'Electronics';`

5.2 — Sorting & Filtering

Query 4: SELECT with ORDER BY:

e.g. `SELECT * FROM products ORDER BY price DESC;`

Query 5: SELECT with LIMIT:

e.g. `SELECT * FROM orders ORDER BY created_at DESC LIMIT 10;`

Query 6: SELECT with LIKE / pattern matching:

e.g. `SELECT * FROM customers WHERE name LIKE 'A%';`

5.3 — Aggregate Functions

Query 7: COUNT, SUM, AVG, MIN, or MAX:

e.g. `SELECT COUNT(*) AS total_orders FROM orders;`

Query 8: GROUP BY with aggregate function:

e.g. `SELECT category, COUNT(*) AS total FROM products GROUP BY category;`

Query 9: GROUP BY with HAVING clause:

e.g. `SELECT customer_id, SUM(amount) AS total FROM orders GROUP BY customer_id HAVING total > 5000;`

5.4 — JOIN Queries

Query 10: INNER JOIN between two tables:

e.g. `SELECT u.name, o.order_date FROM users u INNER JOIN orders o ON u.id = o.user_id;`

Query 11: LEFT JOIN (or RIGHT JOIN):

e.g. `SELECT u.name, o.amount FROM users u LEFT JOIN orders o ON u.id = o.user_id;`

Query 12 (Bonus): Subquery or Nested SELECT:

e.g. `SELECT name FROM users WHERE id IN (SELECT user_id FROM orders WHERE amount > 1000);`

5.5 — Results Screenshot / Output

Paste a screenshot OR write the output of at least 3 queries in the space below.

[Paste Query Output Screenshots Here]

PART 6 — DATABASE BACKUP

You must take a complete backup of your database and attach the .sql dump file with your submission.

6.1 — How to Take a Backup (MySQL)

Follow the steps below to create your database backup using mysqldump.

Step 1	Open Terminal / Command Prompt on your machine.
Step 2	Type the following command (replace 'your_db_name' and 'username'):
Step 3	<code>mysqldump -u username -p your_db_name > project_backup.sql</code>
Step 4	Press Enter. You will be prompted for your MySQL password.
Step 5	A file named project_backup.sql will be created in your current directory.
Step 6	Attach this .sql file with your project folder / ZIP submission.

6.2 — Backup Confirmation

Write the exact command you used to take the backup:

Copy the command from your terminal here.

Write the name of the backup file you are attaching:

e.g. library_db_backup.sql

Describe what the backup file contains (tables, rows of data):

e.g. The backup contains 5 tables: users, orders, products, categories, reviews. Total ~200 rows.

6.3 — Restore Command (Bonus)

Write the command to RESTORE the database from the backup file:

e.g. mysql -u username -p new_db_name < project_backup.sql

PART 7 — REFLECTION & LEARNINGS

What challenges did your group face during this project?

What did you learn from designing and implementing this database?

If you were to improve this project further, what features would you add?

Individual Contribution

Member Name	Role / Contribution	Signature

PART 8 — INSTRUCTOR MARKING RUBRIC

Section	Max Marks	Marks Awarded	Comments
1. Schema Diagram (ERD)	10		
2. Normalization (1NF, 2NF, 3NF)	15		
3. DDL Queries	15		
4. DML Queries	15		
5. DQL Queries (10+ queries)	20		
6. Query Output / Screenshots	5		
7. Database Backup (.sql file)	10		
8. Reflection & Presentation	10		
TOTAL	100		

Instructor Signature:

General Feedback:
